

AP CSP | Lesson 27: Impacts & Digital Divide

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B separates device availability from connection availability. A ignores network access; C invents a bandwidth guarantee; D ignores interface and other access barriers.

Q2. Answer: C

C links one effect to different stakeholder consequences. A excludes benefits; B excludes affected residents and shops; D incorrectly equates unintended with harmful.

Q3. Answer: D

D follows from the stated original intention and allows a possible benefit. A is not implied; B is not supported; C overclaims that this particular use could never have been anticipated.

Q4. Answer: C

C describes geographic and cost-related access differences. A concerns preferences; B naming; D file types. None of A, B or D alone establishes an access disparity.

Q5. Answer: A and D

A is a matched practical intervention; D reflects the scale of access differences. B ignores connection, usability and other barriers. C excludes individual and organizational action.

Q6. Answer: B

B keeps the conclusion within the model. A confuses access with observed participation; C invents equal benefits; D treats assumed costs as measured costs.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Benefit: planners can notice an overlooked stop and direct shade resources to waiting residents. Harm: a misleading report could direct limited resources away from residents with greater need. Both are possible outcomes, not measured effects.

1 mark: Names a stakeholder for the benefit.

1 mark: Explains a plausible map-to-benefit mechanism.

1 mark: Names a stakeholder for the harm.

1 mark: Explains a plausible map-to-harm mechanism; a different supported example is accepted.

Q8. 4 marks

Answer: It is an unintended use relative to the stated planning purpose. It may help residents discover art. Creators could consult affected people about plausible misuse before release. They cannot anticipate every later use.

1 mark: Classifies the art-walk use as unintended relative to the stated purpose.

1 mark: Explains a plausible benefit of that new use.

1 mark: Gives a specific responsible action, such as consultation or misuse testing.

1 mark: States that all future uses or effects cannot be predicted.

Q9. 4 marks

Answer: No action: 40. Connection only: 60. Device only: 55. Both: $85 = 40 + 20 + 15 + 10$; the dual-barrier group joins only after both barriers are removed.

1 mark: Baseline 40.

1 mark: Connection only 60.

1 mark: Device only 55.

1 mark: Both 85 with an explanation that all barriers of the dual-barrier group are removed.

Q10. 4 marks

Answer: Both plans cost 4. Connection+device enables 85; connection+skills+accessibility enables 75. The first leaves skills-only 10 and accessibility-only 5 blocked; the second leaves device-only 15 and dual-barrier 10 blocked. Model totals do not prove actual participation or policy fairness.

1 mark: Gives cost 4 for each feasible plan.

1 mark: Gives access totals 85 and 75 in the correct order.

1 mark: Identifies all remaining groups for both plans.

1 mark: States a relevant limit of using access totals alone.

Q11. 4 marks

Answer: The hotspot can address connection cost or availability nearby. Office hours exclude some working residents. An accessible interface addresses interaction barriers for users who need it. Measure successful submissions across affected groups after the intervention.

1 mark: Identifies connection availability or cost as a barrier the hotspot can address.

1 mark: Explains how office hours limit access for some residents.

1 mark: Matches an accessible interface or another justified action to a stated remaining barrier.

1 mark: Proposes a relevant observable measure beyond merely counting hotspots.

Q12. 4 marks

Answer: Enabled access does not prove that all 100 will submit reports, and participation does not prove equal benefit. A pilot could measure completed submissions and reasons for nonparticipation. If interface difficulty persists, revise the interface and retest.

1 mark: Identifies the unsupported claim that all enabled people will contribute.

1 mark: Identifies the unsupported equal-benefit claim.

1 mark: Proposes a concrete field measure relevant to participation or benefit.

1 mark: Links a possible measurement result to a relevant revision and check.